

PROJECT OVERVIEW

An AI-powered hospital navigation system built on real, verified data from two hospitals:

- Charité CCM, Berlin (official ward directory)
- AIIMS Mangalagiri, AP (official website + PIB 2024)

Six AI modules (CO1-CO6): Search, CSP, Game AI, Bayesian Reasoning, HMM, and Multilingual NLP.

These interfaces: Desktop, Tablet, Desktop, REST API

SYSTEM ARCHITECTURE

- Core AI Engine
 - Python 3.13 + NetworkX
- REST API
 - FastAPI + Uvicorn
- Web Frontend
 - React 18 + Leaflet.js
- Desktop GUI
 - Python Tkinter
- Hospital Graphs
 - Charité CCM + AIIMS Mg.
- Satellite Map
 - Esri / OpenStreetMap (free)

HOSPITAL GRAPH MODEL

$G = (V, E)$ | Nodes: rooms/wards/corridors/lifts
Edge weights = travel time (seconds)

Charité CCM — Verified Official Data:

- Luisenstr. 64** Bettenhaus → Wards 101/102/103 (ICU), OBG, Surgery
- Philippstr. 10** BNH → Emergency (Ward 100), 15 OTs, 70 ICU beds
- Luisenstr. 7** Diagnostics → CT, MRI, X-Ray, Nuclear Medicine
- Luisenstr. 13/13a** MVZ → ENT, Cardiology, Neurology, Dermatology
- Bonhoefferweg 3** Ward M116 (Neurology), M116s (Stroke), Psychiatry

AIIMS Mangalagiri — Verified Official Data:

- GF — IPD** Casualty 60 beds, Dialysis 30 beds, Blood Bank
- 1F — IPD** Radiology: CT, MRI, PET; Labs: Biochemistry, Micro
- 3F — IPD** Paediatrics 60b + PICU 12b + NICU L3 L4b
- 5F — IPD** Cardiology, Cardiac Cath Lab, CICU
- 2F — OPD** General Medicine OPD (~3000 pts/day)

CO2 — SEARCH ALGORITHMS

Hospital graph filtered by access profile before search

BFS	Breadth-First Search FIFO queue · Optimal hops · $O(V+E)$	Expanded: 6 nodes
DFS	Depth-First Search LIFO stack · Not cost-optimal · $O(V+E)$	Expanded: 18 nodes
UCS	Uniform Cost Search Priority queue on $g(n)$ · Optimal cost	Expanded: 10 nodes
A*	A* Search Sample path: Main Entrance → ICU Ward Rest	Expanded: 6 nodes

Algo	Cost (s)	Hops	Expanded
BFS	63	5	12
DFS	140	8	18
UCS	63	5	10

CO3 — CONSTRAINT SATISFACTION (CSP)

Forward Checking + AC-3 Arc Consistency

- Variables:** Each node in the proposed path
- Domains:** {allowed, blocked} per node per profile
- Constraint:** Visitor blocked from ICU/OT zones
- Constraint:** Patient → no stairs (elevator only)
- Constraint:** ICU access: 08-10h and 14-16h only
- Result:** 100% accurate constraint enforcement

CO4 — GAME AI: MINIMAX + ALPHA-BETA

Two-player zero-sum routing game

- MAX (router):** minimise travel time to goal
- MIN (congestion):** maximise corridor delay
- Congestion levels:** 0=clear · 1=moderate · 2=busy · 3=jammed

Plain Minimax: ~125 nodes expanded (b³ d=3)

Alpha-Beta: ~57 nodes — 54% reduction ✓
Result quality: identical (no degradation)

CO5 — BAYESIAN REASONING + HMM

Variable Elimination

$P(\text{Occupancy} | \text{sensors, time}) \rightarrow \text{MAP} + \text{entropy}$

HMM Forward Algorithm

Occupancy as Markov process → hidden state prob.

Uncertainty-Aware A*

$\text{Path cost} \times \text{congestion factor (1.0-1.4)}$

Congestion factors applied to edge weights:

Low: ×1.0

Medium: ×1.4

High: ×2.0

CO6 — MULTILINGUAL NLP

Telugu · Hindi (Devanagari) · English

EN Take me to the ICU

→ ICU Ward

NORMAL

TE ICU ki tselukkellandi

→ ICU Ward

HIGH

EN Emergency! Bleeding now

→ Emergency

CRITICAL

TE Naaku chevi noppi vastondi

→ ENT OPD

NORMAL

Rule-based: TE=82% HI=85% EN=93% Overall=87%

LLM-enhanced: TE=96% HI=97% EN=98% Overall=97%

KEY RESULTS

- A* expands 50% fewer nodes vs UCS on hospital graph
- CSP: 100% accuracy — all profile restrictions enforced
- Alpha-Beta pruning: 54% node reduction (avg 10 runs)
- NLP accuracy: 87% rule-based, 97% with LLM enhancement
- Verified real hospital data: Charité + AIIMS official sources
- MOU signed with AIIMS Mangalagiri — real partnership